

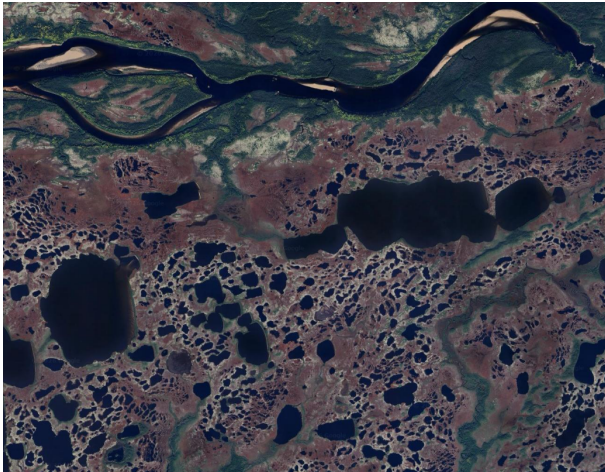


Metapopulation dynamics

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Exercise



Spatially implicit metapopulation dynamics

```
DEFINE vector of J patches
FOR time steps
  Compute occupancy p
  FOR patch x
    IF presence[x] = 0
      IF rand < cp
        presence[x] = 1
    ELSE rand < e
      presence[x] = 0
```

1. Run simulations with $e = 0.2$ and $c = 0.3$
2. What happens with $J = 1000$, $J = 100$, $J = 10$?

Theory



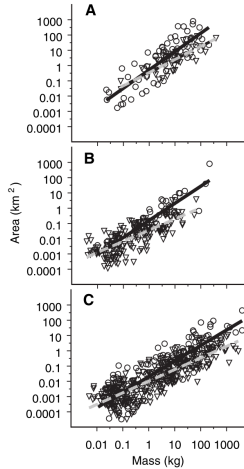
Patch

Definition : a discrete area of suitable habitat that can support a local breeding population, separated from other similar areas by unsuitable matrix territory where individuals cannot permanently live





Scale-dependence



Jetz et al. 2004. Science.

Patch Colonization





Patch Colonization



- ▶ Dispersal
- ▶ Find empty patches
- ▶ Establishment
- ▶ Population growth



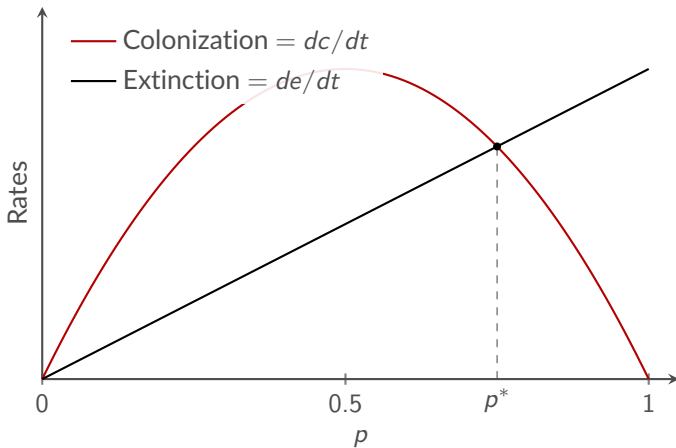
Patch Extinction



- ▶ Demographic stochasticity
- ▶ Selection
- ▶ Disturbance



Combining processes





Levins model

The Levins metapopulation model describes the proportion p of occupied patches in a landscape as :

$$\frac{dp}{dt} = \underbrace{cp}_{\text{propagules}} \underbrace{(1-p)}_{\text{empty patches}} - \underbrace{ep}_{\text{extinction}}$$



Levins model Persistence

The fundamental result of metapopulation theory is that persistence requires :

$$c > e$$



Levins model Equilibrium

Time dynamics follow asymptotic curve :

$$p(t) = \frac{1}{\frac{1}{p_0} \exp(-(c - e)t) + \frac{c}{c - e} (1 - \exp(-(c - e)t))}$$

With equilibrium :

$$p^* = 1 - \frac{e}{c}$$



Levins model Interpretation

- ▶ Occupancy increases non-linearly with colonization
- ▶ Occupancy decreases linearly with extinction
- ▶ Dynamics are globally stable
- ▶ Asymptotic growth
- ▶ Variability depends on the number of patches



Levins model Assumptions

- ▶ Discrete landscape
- ▶ Separation of time scales
- ▶ Infinite number of patches
- ▶ All patches are identical
- ▶ Landscape is constant over time
- ▶ Surrounding matrix is inhospitable
- ▶ Independence of c and e
- ▶ Global dispersal

Exercise



Spatially explicit dynamics

```
DEFINE presence      # vector of J patches
DEFINE XY            # array of spatial coordinates
DEFINE ConMat        # Connectivity matrix
DEFINE Nmax          # Number of nearest neighbors
FOR time steps
  FOR patch x
    IF presence[x] = 0
      ConPop = ConMat%*%presence / Nmax
      IF rand < cp
        presence[x] = 1
    ELSE rand < e
      presence[x] = 0
```



Spatially explicit dynamics Questions

1. Run simulations with $e = 0.2$ and $c = 0.3$
2. What happens with $J = 1000$, $J = 100$, $J = 10$?
3. **Advanced** : compare with random connectivity

Back to theory



Pair approximation

- ▶ Consider patches are aligned along a circle
- ▶ Dispersal only occurs between adjacent patches
- ▶ We model the dynamics of pairs of patches



Pair approximation Model

We have a set of three differential equations :

$$\frac{dp_{EE}}{dt} = 2ep_{OE} - cp_{EE} \left(\frac{p_{OE}}{p_{OE} + p_{OO}} \right)$$

$$\frac{dp_{OE}}{dt} = \frac{c}{2} p_{EE} \left(\frac{p_{OE}}{p_{OE} + p_{OO}} \right) + ep_{OO} - \left[e + \frac{c}{2} \left(1 + \frac{p_{OE}}{p_{OE} + p_{OO}} \right) \right] p_{OE}$$

$$\frac{dp_{OO}}{dt} = c \left(1 + \frac{p_{OE}}{p_{OE} + p_{OO}} \right) p_{OE} - 2ep_{OO}$$



Pair approximation Solution

Which yields the equilibrium metapopulation occupancy :

$$p = \frac{1 - 2(e/c)}{1 - (e/c)}$$



Pair approximation Autocorrelation

Local dispersal has for result that pairs of patches tend to be more similar than with global dispersal.

The spatial autocorrelation in this model is :

$$\rho = \frac{e}{c}$$

And we can re-arrange equilibrium solution as :

$$p^* = 1 - \frac{e}{c} \frac{1}{1 - \rho}$$



Pair approximation Interpretation

- ▶ Spatially explicit dispersal generates positive autocorrelation
- ▶ Autocorrelation declines with increasing colonization
- ▶ Spatially explicit dispersal reduces equilibrium occupancy

The [very] important notion of dispersal limitation becomes clear : ecological conditions may result in situations where a species is absent from a location where it should be.



Habitat destruction

The standard Levins model assumes that the entire set of suitable patches is available for colonization.

Habitat destruction D can be introduced as follows :

$$\frac{dp}{dt} = cp(1 - D - p) - ep$$



Habitat destruction

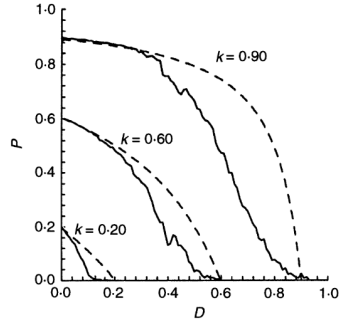
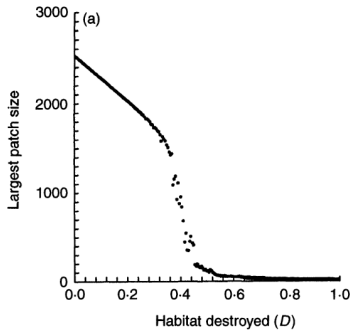
Persistence now requires that :

$$c(1 - D) > e$$

Which means that a species may go extinct, even if $c > e$ and that there are remaining suitable habitats.



Habitat destruction **Spatially explicit**



Spatially explicit dynamics magnifies the effect of habitat destruction